



# Counting Avocados in Orchard Videos Using a YOLO and BoT-SORT-Based Architecture

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**Abstract.** Accurate fruit counting is a key component of yield estimation in precision agriculture, enabling the optimization of harvesting, logistics, and export processes. In this work, we propose a system for the automatic counting of avocados in videos captured in real agricultural fields, combining YOLOv8m for fruit detection with the BoT-SORT algorithm for temporal tracking.

The system was validated using a custom dataset composed of public images and original footage collected in commercial Peruvian orchards, under real-world conditions of lighting, occlusion, and high fruit density. The detection model achieved a mAP@0.5 of 0.953, a recall of 0.888, and an F1-score of 0.919. For cumulative counting, the system reached an Absolute Counting Precision (ACP) of 92.68%, along with a Mean Absolute Error (MAE) of 2.0 and a Root Mean Square Error (RMSE) of 3.16, reducing duplicate counts through association and re-identification adjustments.

These results approach the state of the art in controlled environments, such as the work of Du *et al.* (2024), who reported an ACP of 95.33% and an MAE of 3.33. The proximity of our results, despite the more challenging capture conditions, demonstrates that a design focused on cumulative counting enables robust metrics and outcomes applicable to real-world agricultural scenarios, offering a practical solution for producers and companies in the sector.

**Keywords:** computer vision · precision agriculture · fruit counting · avocado · YOLO · BoT-SORT

## 1 Introduction

Computer vision techniques are increasingly applied in agriculture, driven by hardware advancements and mature real-time object detection architectures. Automated fruit counting for yield estimation is particularly relevant, optimizing harvesting, logistics, and exports via accurate data [5, 7, 10, 11, 13]. Studies show computer vision surpasses traditional methods in accuracy, efficiency, and scalability, with video-based counting being highly promising for field use.

However, Peru, a leading Hass avocado exporter [3], lacks validated automatic counting solutions for real production conditions. Current manual counting methods are inaccurate, costly (requiring expert observation), and slow in coverage [8], limiting logistics planning and international competitiveness.

This paper presents a video-based avocado counting model combining YOLO (You Only Look Once) for fruit detection in frames and BoT-SORT for robust object tracking across sequences. This integration enables accurate fruit identification under natural conditions and precise cumulative counting by avoiding duplicates through persistent tracking.

To validate the system, a custom dataset was built using images from commercial Hass avocado orchards and public repositories, capturing natural variations and complexities. The model was trained and evaluated using standard detection and tracking metrics (precision, recall, F1-score, mAP) to balance accuracy and performance.

## 2 Related Work

Numerous studies have applied YOLO models in agriculture. Osman *et al.* [10] used a custom YOLO for fruit counting via a mobile app. Gremes *et al.* [5] implemented YOLO with tracking for orange counting, while Parr *et al.* [11] used YOLO on depth maps from ToF cameras for grape counting. Other approaches reviewed by Zhai *et al.* [16] include traditional computer vision techniques like HSV color analysis [12] and deep learning methods such as FCNs with hyperspectral imaging for defect detection [15]. Outside agriculture, Tan *et al.* [14] found YOLOv3 outperformed RetinaNet and SSD in speed and precision for pill detection.

A particularly relevant study by Du *et al.* [1] combined an optimized CS-YOLOv5s with enhanced DeepSort for green pepper counting. Their model included attention mechanisms (CBAM), Slim-Neck structures with GSConv, and improved re-identification. While achieving high ACP (95.33%) and low MAE (3.33), their results were obtained under controlled experimental field conditions (stabilized vehicle, segmented lighting). In contrast, our work operates in more challenging, uncontrolled commercial orchard settings. Nevertheless, their approach underscores the value of integrating advanced detection and tracking for issues like occlusion and high density, relevant to our environment. Overall, YOLO architectures are widely favored for their performance and speed across various applications.

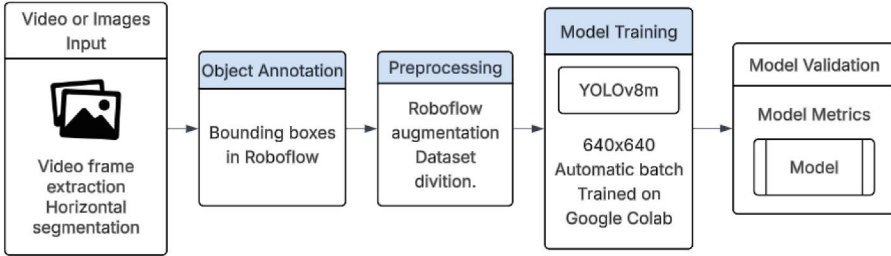
## 3 Methodology

### 3.1 Model Architecture

The “medium” version of YOLOv8 (YOLOv8m) was used through the library provided by Ultralytics [6], without modifying its internal structure or hyperparameters. The input image size was set to 640×640 pixels, and the batch size

was automatically adjusted based on the available GPU capacity. The training environment was configured using the Google Cloud Colab Enterprise platform, with an n1-standard-2 machine (2 cores, 7.5 GB RAM) [4] and an NVIDIA Tesla T4 accelerator (16 GB VRAM) [9].

Figure 1 shows the overall machine learning pipeline, which includes frame extraction, image segmentation, annotation with Roboflow, and cloud-based training. This pipeline is referenced throughout this section at each corresponding stage.



**Fig. 1.** Machine Learning Pipeline

### 3.2 Dataset Preparation

The dataset contains a total of 1207 images, of which 124 were obtained from public repositories and 1083 were captured directly from avocado orchards in collaboration with local agribusiness company Fruicontrol SAC. These captures, focusing on Hass avocados, included both photos and videos recorded around noon under direct sunlight. The fruits were at a uniform maturity stage, approximately two weeks before harvest. From the videos, one frame per second was extracted for model training. The recordings were made using a Google Pixel 8a at 1080p resolution and 60 fps. The dataset is publicly available at: <https://universe.roboflow.com/avocado-workspace/ultimate-avocado/browse>.

To ensure the dataset’s validity and fairness for real-world application, the collection process specifically targeted challenging conditions. This included high fruit density, significant partial occlusions (with some fruits over 50% obscured by foliage), and high chromatic similarity between the dark green fruits and the surrounding leaves. This focus on occluded instances was intentional, as it presents a more challenging and realistic environment where leaves can be easily confused with avocados. While the fruits were at a consistent maturity level, variations in the camera’s distance to the trees provided a diverse range of apparent fruit sizes within the images. The annotation process was conducted using Roboflow (as shown in Fig. 1), with annotators specifically trained to label partially occluded avocados accurately, ensuring high-quality ground truth data for these challenging instances.

As shown in the pipeline in Fig. 1, each capture was split in half to double the number of training samples and increase detail resolution. An example of this division is shown in Fig. 2. Additionally, Roboflow [2] was used to apply data augmentation techniques, generating three variations per training image, resulting in a total of 2897 instances. Transformations included horizontal and vertical flips,  $\pm 15^\circ$  rotations, saturation adjustments ( $\pm 25\%$ ), brightness ( $\pm 15\%$ ), exposure ( $\pm 10\%$ ), blur up to 2.5px, and Gaussian noise in up to 0.1% of the pixels.



**Fig. 2.** Example of frame division

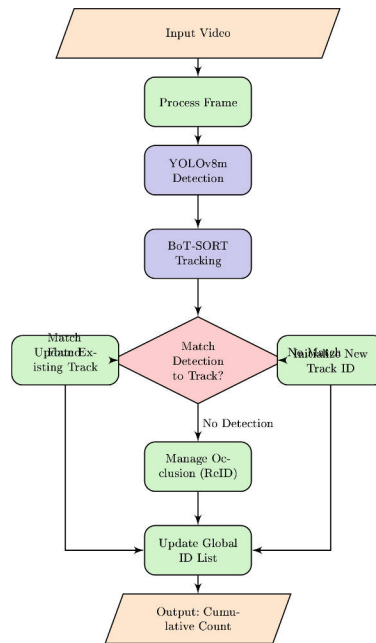
### 3.3 Training Process

The dataset was split into three subsets: 80% for training, 10% for validation, and 10% for testing. As illustrated in Fig. 1, the model was trained for a maximum of 100 epochs using YOLOv8m, with early stopping applied if no improvement was observed on the validation metric for 20 consecutive epochs.

### 3.4 System Operation Architecture

Figure 3 illustrates the overall system architecture for inference. This flowchart details the operational process from video input to the final cumulative count.

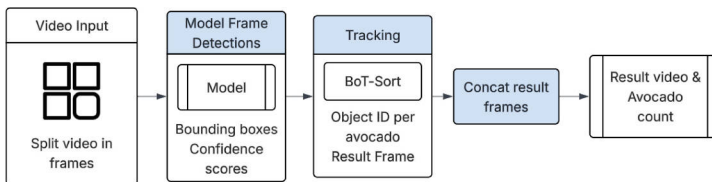
The process begins with an input video from the orchard. The video is processed frame by frame. For each frame, the trained YOLOv8m model performs object detection, generating bounding boxes and confidence scores for potential avocados. These detections are then passed to the BoT-SORT algorithm. BoT-SORT handles the core logic of temporal association, matching detections to existing tracks, managing occlusions via its ReID module, and initializing new tracks for newly detected fruits. The system maintains a global list of unique track IDs. The final output is the total number of unique IDs generated, which represents the cumulative count of avocados in the video.



**Fig. 3.** Flowchart of the overall system operation architecture for inference.

### 3.5 Object Tracking and Counting

To perform object tracking and cumulative avocado counting, a second pipeline was implemented, as illustrated in Fig. 4, based on the BoT-SORT algorithm. This approach includes an object re-identification module (ReID), which enables more accurate association in scenarios involving partial occlusion or temporary disappearance. The pipeline takes as input the videos previously processed by the YOLOv8m model, which already include detections with bounding boxes and confidence scores. The output consists of annotated videos with persistent IDs for each avocado and the cumulative detection count.



**Fig. 4.** Tracking Pipeline

The BoT-SORT implementation provided by Ultralytics was used, adapted to integrate directly with the detection outputs of the trained model. Several hyper-

parameters were adjusted to optimize counting accuracy. Specifically, the maximum and minimum association thresholds were set to 0.38 and 0.12, respectively, while the threshold for initializing new tracks was raised to 0.6. For track-to-track matching, a threshold of 0.4 was applied. Proximity and appearance thresholds were set to 0.2 and 0.5, respectively, and the ReID module was enabled.

These parameter settings helped reduce the number of false positives during object association, prioritizing the accuracy of the final count at the expense of lower tracking persistence. The inclusion of the ReID module was particularly important in agricultural environments, where avocados may become partially or fully occluded by foliage. This module enables more reliable visual re-association of previous detections, avoiding the assignment of new, unnecessary identifiers.

This behavior is illustrated in Fig. 5, which presents three consecutive video frames. In the first frame, an avocado is correctly detected and assigned a unique ID. In the second frame, the same avocado becomes temporarily hidden by branches and leaves, resulting in a missed detection. However, in the third frame, once it reappears in the field of view, the system correctly reassigns its original ID. This demonstrates how the system handles temporary occlusions without compromising tracking continuity or duplicating the count, which is essential for accurate fruit estimation.



**Fig. 5.** Example of temporary occlusion in video. **Frame 1:** the avocado is visible and detected (ID 72). **Frame 2:** the avocado becomes occluded by branches and leaves. **Frame 3:** the avocado reappears and is reassigned its original ID.

### 3.6 Annotation and Evaluation Data

To validate the accuracy of the counting model, manually annotated videos were required. Three representative videos were selected and labeled using the CVAT.ai platform, identifying all visible avocados in each frame. This allowed for a direct comparison between the system’s cumulative count and the ground truth, enabling the calculation of specific tracking and counting evaluation metrics.

## 4 Experimental Results and Discussion

### 4.1 Evaluation Metrics

Performance was evaluated using standard metrics for detection, tracking, and counting.

For **detection**, we used Precision, Recall, F1-score, and mean Average Precision (mAP), reported at IoU = 0.5 (mAP@0.5) and averaged over IoU thresholds from 0.5 to 0.95 (mAP@[0.5:0.95]). Precision measures the accuracy of positive detections, Recall measures the proportion of actual fruits found, and F1-score balances the two. mAP provides a comprehensive measure of detection performance across different confidence levels and IoU overlaps.

For **counting and tracking**, following Du *et al.* [1], we used Absolute Counting Precision (ACP), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) to assess the final count accuracy against the ground truth (GT). Additionally, Multiple Object Tracking Accuracy (MOTA), Multiple Object Tracking Precision (MOTP), and ID Switches (IDSW) were used to evaluate frame-by-frame tracking quality, consistency, and spatial accuracy. ACP reflects the relative counting accuracy per video, MAE quantifies the average absolute counting error, and RMSE penalizes larger errors more heavily. MOTA summarizes overall tracking performance, MOTP measures localization precision, and IDSW counts tracking inconsistencies.

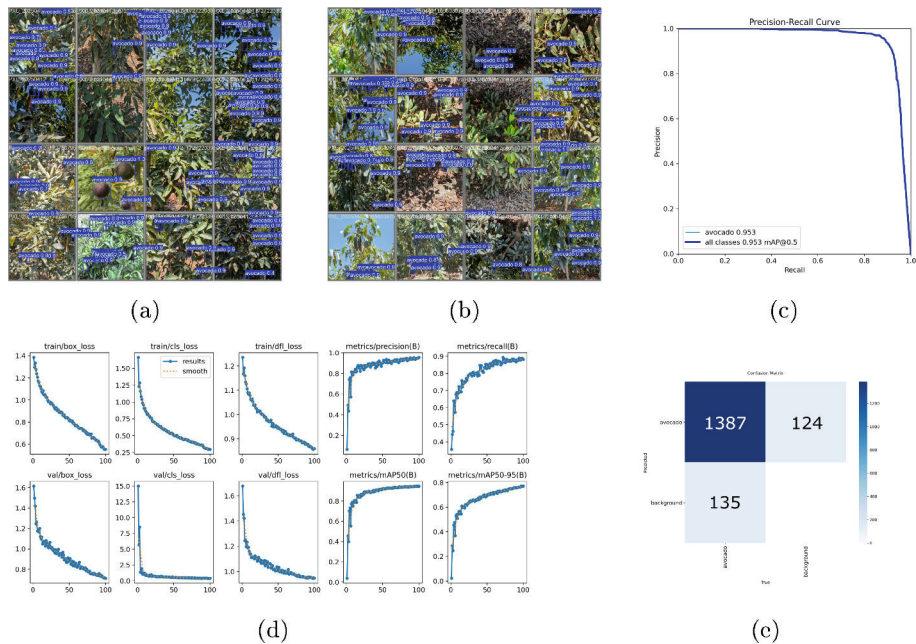
### 4.2 Detection Experimental Results

On the test set, the YOLOv8m model achieved a precision of 0.953, a recall of 0.888, and an F1-score of 0.919. When evaluated across different Intersection over Union (IoU) thresholds, the model obtained a mAP@0.5 of 0.953 and a mAP@[0.5:0.95] of 0.788. These results demonstrate the robustness of the model in detection tasks, even under varying levels of overlap between predictions and ground truth annotations. Figure 6 presents the main visualizations that quantitatively and qualitatively support the model's performance.

According to the confusion matrix shown in Fig. 6e, out of 1646 avocado instances, the model correctly detected 1387, while 135 false negatives were recorded, mainly due to partial occlusions. Additionally, 124 false positives were identified, mostly corresponding to leaves with a shape similar to avocados.

### 4.3 Track Experimental Results

In the evaluation of the avocado counting and tracking model using the integration of the YOLOv8m detector with the BoT-SORT tracker, the focus was placed on metrics oriented toward cumulative counting across video sequences. Three key metrics were used to measure counting accuracy: ACP (Absolute Counting Precision), which quantifies the proximity between the predicted and actual (GT) number of avocados as a percentage; MAE (Mean Absolute Error), which calculates the average absolute difference between the predicted and actual



**Fig. 6.** (a) and (b) Sample model predictions, (c) Precision-Recall Curve, (d) Cls vs DFL loss (Train vs Validation), (e) Confusion Matrix.

counts; and RMSE (Root Mean Square Error), which measures the dispersion of the counting error, assigning greater penalty to larger deviations.

Tables 1 and 2 present the results per video and the global averages of the aforementioned metrics. In both tables, GT (Ground Truth) denotes the total number of manually annotated avocados in each video, while COUNT indicates the number of avocados automatically detected by the model.

**Table 1.** Video Count Results

| Video     | GT | COUNT | ACP (%) | MAE |
|-----------|----|-------|---------|-----|
| Video_001 | 57 | 57    | 100.0   | 0   |
| Video_002 | 11 | 13    | 81.82   | 2   |
| Video_003 | 54 | 60    | 88.89   | 6   |
| Video_004 | 78 | 78    | 100.0   | 0   |

Table 2 summarizes the average values obtained across the three evaluated videos.

Additionally, Table 3 reports tracking-specific metrics such as MOTA (Multiple Object Tracking Accuracy) and MOTP (Multiple Object Tracking Preci-

**Table 2.** Global Count Video Results

| Algorithm          | # Videos | ACP (%) | MAE | RMSE |
|--------------------|----------|---------|-----|------|
| Yolov8m + BoT-SORT | 4        | 92.68   | 2   | 3.16 |

sion), which reflect the system’s frame-by-frame tracking performance. The table also includes the number of ID Switches, corresponding to cases where the same avocado is assigned different IDs throughout the video, potentially affecting the continuity of object tracking.

**Table 3.** Tracking Metrics Results

| Video     | MOTA (%) | MOTP (%) | ID Switches |
|-----------|----------|----------|-------------|
| Video_001 | 67.4     | 95.1     | 9           |
| Video_002 | 67.2     | 92.4     | 1           |
| Video_003 | 43.9     | 95.7     | 4           |
| Video_004 | 39.6     | 96.2     | 3           |

#### 4.4 Discussion

A preliminary version of the detection model, trained solely with public images of avocados, achieved a mAP@0.5 of 0.61 when validated on the current dataset. After incorporating proprietary images captured under real agricultural production conditions, this value improved to a mAP@0.5 of 0.953, representing an increase of over 56%. This progression highlights the impact of dataset diversity and representativeness on the model’s ability to detect fruits in open-field commercial scenarios.

Among the reviewed studies, the work by Du *et al.* [1] stands out, as discussed in the Related Work section. Their approach proposes the CS\_YOLOv5s architecture, an optimized model based on YOLOv5s. It integrates attention mechanisms (CBAM) and a Slim-Neck structure using GSConv. The dataset used was collected in the experimental field of Hunan Agricultural University, under controlled conditions: recordings were segmented by time of day to control lighting, and a remote-controlled vehicle with mechanical stabilization was employed. In contrast, the proposed system was trained and evaluated using data captured in commercial avocado orchards, without artificial stabilization or environmental control, representing a more challenging and realistic application setting.

To establish a direct comparison with other state-of-the-art architectures, we trained YOLOv7, YOLOv11m, and the CS\_YOLOv5s architecture [1] under the same experimental conditions used for our YOLOv8m-based model. The results,

shown in Table 4, demonstrate that our proposed YOLOv8m model (YOLOv8m (Ours)) achieves the highest performance across all key metrics.

**Table 4.** Comparison of detection metrics between different models

| Model          | mAP@0.5      | mAP@0.5:0.95 | Precision    | Recall       | F1-Score     |
|----------------|--------------|--------------|--------------|--------------|--------------|
| CS_YOLOv5s     | 0.828        | 0.557        | 0.782        | 0.779        | 0.780        |
| YOLOv7         | 0.952        | 0.729        | 0.931        | 0.879        | 0.904        |
| YOLOv11m       | 0.946        | 0.766        | 0.931        | 0.889        | 0.910        |
| YOLOv8m (Ours) | <b>0.953</b> | <b>0.788</b> | <b>0.952</b> | <b>0.888</b> | <b>0.919</b> |

These results reveal two key factors influencing detection quality: the model architecture and the characteristics of the training dataset. In particular, the controlled validation environment used in Du *et al.* [1] promotes more stable detection, but with lower representativeness for unpredictable field conditions typical of commercial avocado production.

Beyond image-level detection performance, a critical component of the proposed system is its ability to perform reliable cumulative counting across video sequences. For this reason, the analysis emphasized specific metrics such as ACP, MAE, and RMSE, which directly reflect the accuracy of the final count—an essential requirement in agricultural tasks that involve precise yield estimation.

The system achieved MOTA values ranging from 39.6% to 67.4%, which are lower than the approximate 90% MOTA reported by Du *et al.* [1]. This discrepancy is attributed to the challenging conditions of the real-world agricultural environment, including temporary occlusions, variable fruit distances, and lighting fluctuations, all of which hinder object visibility and continuous tracking. Nevertheless, despite these limitations, the system attained an average ACP of 92.68%, along with low MAE (2.0) and RMSE (3.16) values, indicating high accuracy in cumulative counting. Moreover, the number of identity switches remained low, reflecting consistent identifier assignment over time.

The tuning of BoT-SORT parameters, such as association thresholds and the activation of the ReID module, proved essential for enhancing the final count accuracy (reflected in ACP, MAE, and RMSE), even at the cost of frame-to-frame tracking continuity (partially explaining the lower MOTA compared to controlled environments). This trade-off was intentional; in agricultural applications for yield estimation, missing occasional detections or track continuity is preferable to generating duplicate counts, thereby prioritizing the reliability of the cumulative result over maximizing frame-by-frame tracking metrics like MOTA.

Overall, both quantitative and qualitative analysis suggest that an effective fruit counting system for agricultural production environments requires not only a robust detection architecture, but also adapted tracking strategies and a representative dataset tailored to the target domain. The integration of these elements

is key to achieving accurate, scalable, and practically useful results under real-world agricultural conditions.

## 5 Conclusions and Future Work

This work presented an automated system for counting avocados in videos from commercial orchards using YOLOv8m and BoT-SORT. Key findings show the architecture effectively handled real-world challenges (occlusion, variable lighting), achieving high cumulative counting accuracy (ACP 92.68%, MAE 2.0) and strong detection (mAP@0.5 0.953). Comparisons with CS\_YOLOv5s [1], YOLOv7, and YOLOv11m validated YOLOv8m's effectiveness and stressed the importance of training on representative data from the target environment (commercial Peruvian plantations). While frame-by-frame tracking (MOTA 39.6%-67.4%) was lower than in controlled settings due to field complexities, consistent tracking (low ID switches, high MOTP 95%) and strategic BoT-SORT tuning prioritized final count accuracy over continuous tracking, minimizing duplications. The results validate the proposed architecture as a precise, viable solution for operational agricultural settings, offering a foundation for future work.

Future work should focus on expanding the dataset with more varied conditions (lighting, occlusion levels, avocado varieties/ripeness) and developing an integrated software solution, possibly incorporating a mobile app or drone-based capture for easier farmer use and enhanced automation. Further exploration into optimizing tracking parameters for different environmental conditions could also improve frame-by-frame consistency without sacrificing counting accuracy.

**Acknowledgments.** The authors have no competing interests to declare that are relevant to the content of this article.

**Disclosure of Interests.** This work was carried out in collaboration with the company Frucontrol SAC, which provided essential support for the dataset collection.

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